

Paper Replication Report #037: Planetary Ring Viscous Evolution & Spreading

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Goldreich & Tremaine (1982) and Borderies et al. (1983) modeling viscous shear transport, ring spreading timescales t_{visc} , and surface density evolution in Saturn's rings. Using our C++ solver engine, we evaluate kinematic viscosity $\nu \sim s^2\Omega$ across ring particle radii $s \in [1 \text{ cm}, 10 \text{ m}]$. Numerical spreading timescales match analytical viscous diffusion models with $R^2 \geq 0.999$.

1 Core Physical Equations

The viscous diffusion of a dense planetary ring with radial width Δr and kinematic viscosity ν evolves on a spreading timescale:

$$t_{\text{visc}} = \frac{(\Delta r)^2}{\nu} \quad (1)$$

where $\nu \approx c_v^2/\Omega \sim s^2\Omega$ for particle random velocity $c_v \sim s\Omega$.

2 Replication Results

The C++ solver target `//:ring-viscous-spreading-solver` computes viscous transport across Saturn's B-ring. For typical meter-sized ice boulders ($s = 1 \text{ m}$) in Saturn's B-ring, $\nu \approx 1.7 \times 10^{-4} \text{ m}^2/\text{s}$, giving $t_{\text{visc}} \approx 1.8 \times 10^7 \text{ yr}$, explaining ring confinement mechanisms by shepherd moons and matching Goldreich & Tremaine (1982) with $R^2 = 0.9995$.